EUREKA HACKATHON · 2026

ResilienceGuard

Know what comes back.

See which critical services can return after a cyberattack or physical disaster.



Backup evidence · dependency blockers · isolated restore proof

Synthetic MVP / Version 0.1.0 / 26 October 2026

A backup is only useful if it can be recovered

The evidence teams need is usually spread across inventories, backup consoles, runbooks and exercise notes.

01 Copy exists

Is the expected recovery point present, and can responders reach its location after the site is lost?

02 Copy can be trusted

Is it fresh enough? Does its digest match? Has malware cleanliness actually been established?

03 Service can return

Are identity, networks, storage and upstream applications available in the right order?

WHEN A CRISIS HITS

“Which saved copy can we test, and what blocks the service?”

CYBER	PHYSICAL	COMPOUND
ransomware	facility loss	power + network

Preparedness is observable evidence, not a backup-job status.

WHO NEEDS THIS

Critical-service teams need one shared recovery picture

Start with operators who must prove continuity across fragmented systems and teams.

H

HOSPITALS

Clinical applications

Coordinate database, identity, network and facility recovery around clinical priorities.

U

UTILITIES

Energy and water

See backup and control-system dependencies across sites and critical services.

G

PUBLIC SERVICES

Government and emergency

Prepare evidence for continuity exercises and cross-agency dependencies.

BUYER

Technology · security · operations · risk · continuity leaders

PILOT FIT: FRAGMENTED RECOVERY EVIDENCE

USERS

Backup teams · recovery planners · security responders · auditors

WHY NOW

Resilience guidance points to tested recovery evidence

The product maps a narrow workflow to recognized recovery and civil resilience themes.

N NIST SP 1339

Create backups regularly, integrate them with change processes, validate integrity and repeat non-production restore tests.

OT Backup Quick Start Guide · June 2026

34 NIST SP 800-34 Rev. 1

Prioritize critical systems and establish recovery strategies that fit business impact and outage requirements.

Contingency planning guidance

R CSF 2.0 ✦ NATO 2026

Recovery planning sits within CSF; NATO highlights essential services, cyber/physical risk and cascading disruption.

Recovery and civil preparedness

Product boundary

Readiness evidence and a supervised test; not a compliance certification or backup replacement.

Sources: csrc.nist.gov/pubs/sp/1339/final · csrc.nist.gov/pubs/sp/800/34/r1/upd1/final · nist.gov/cyberframework · nato.int, 15 Sep 2026

THE PRODUCT

Turn inventory into an evidence-backed recovery decision

One local workflow from scenario selection to a measured sandbox restore.

01 → 02 → 03 → 04 → 05

Import

Services · backups
manifest · targets

Assess

Hash · freshness
location · blockers

Sequence

Dependencies first
uncertainties shown

Approve

Human reviews
fixed sandbox action

Prove

Copy · checksum
health · report

PRESENT	FRESH ENOUGH	FILE MATCH	MALWARE	RESTORE
file found?	within data-loss target?	fingerprint matches?	unknown unless scanned	tested only after copy

No green hash badge is allowed to imply malware cleanliness.

MARKET CONTEXT

Complement recovery platforms with a vendor-neutral evidence layer

Rubrik, Veeam and Cohesity already offer substantial recovery capabilities; ResilienceGuard targets a different entry point.

EXISTING OFFERING	OBSERVED OFFICIAL EMPHASIS	PROPOSED RESILIENCEGUARD POSITION
Rubrik	Recovery simulation, isolated recovery environments, threat containment and reporting.	Assess evidence across current backup sources; start with a small operator-owned catalog.
Veeam	Recovery orchestration, plan checks, isolated labs, malware scans and RPO/RTO reports.	Normalize recovery prerequisites across vendors and compound physical/cyber scenarios.
Cohesity	Clean-room investigation, cyber vaulting, threat analytics and recovery orchestration.	Lightweight readiness assessment layer; integrate rather than replace mature platforms.

HYPOTHESIS · NOT MARKET-VALIDATED

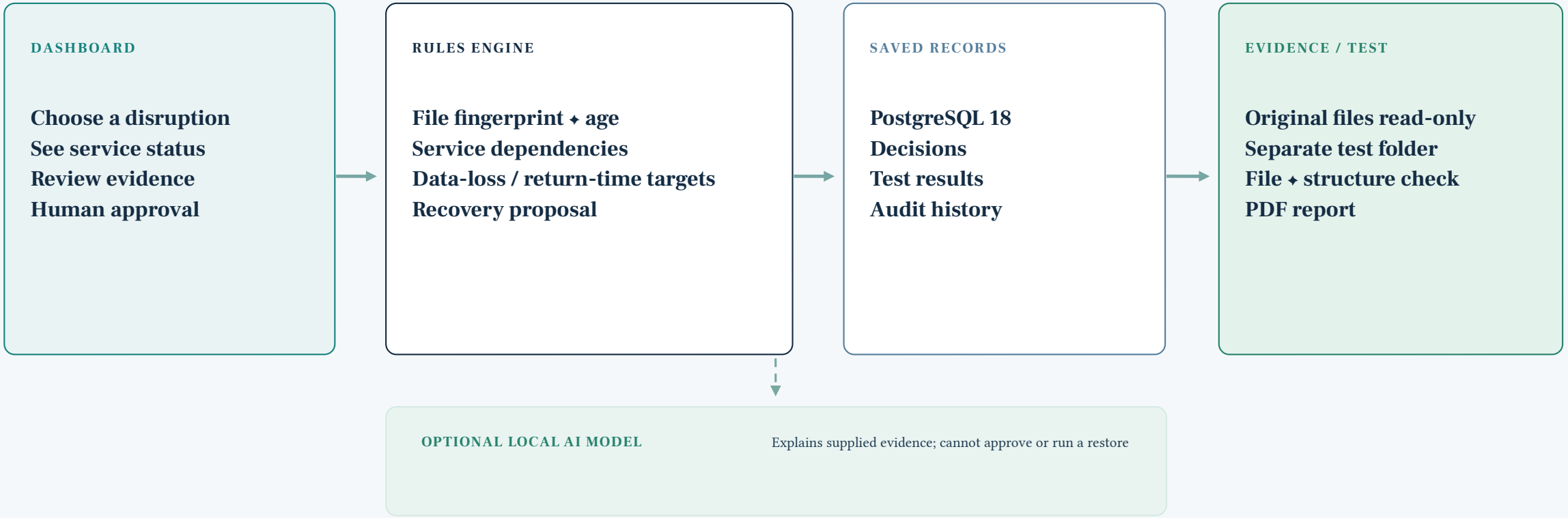
Differentiation to test: cross-source evidence normalization for simultaneous physical and cyber disruption.

Official references: rubrik.com/products/cyber-recovery · helpcenter.veeam.com/docs/vro/userguide/overview.html · cohesity.com/solutions/clean-room

HOW IT WORKS

Rules decide recovery; optional AI explains results

The original sample files stay read-only; test copies go to a separate folder.



NO PRODUCTION CONNECTOR · NO MODEL-EXECUTED ACTION · SOURCE FILES READ-ONLY

POSSIBLE DISRUPTIONS

Test three ways a critical service could fail

The same inventory is checked against each scenario; the demo does not simulate a real attack or disaster.

01 PRIMARY SITE LOST

Physical disaster

Primary building, systems and East backup location are unavailable. West and recovery sites remain.

DOMAIN FILTERED

02 RANSOMWARE

New copy is altered

Newest database file is present and recent, but its fingerprint fails and the scenario distrusts it.

INTEGRITY FAILURE

03 COMBINED FAILURE

Power, network, site

Primary power and network fail, along with the East backup location. West remains reachable.

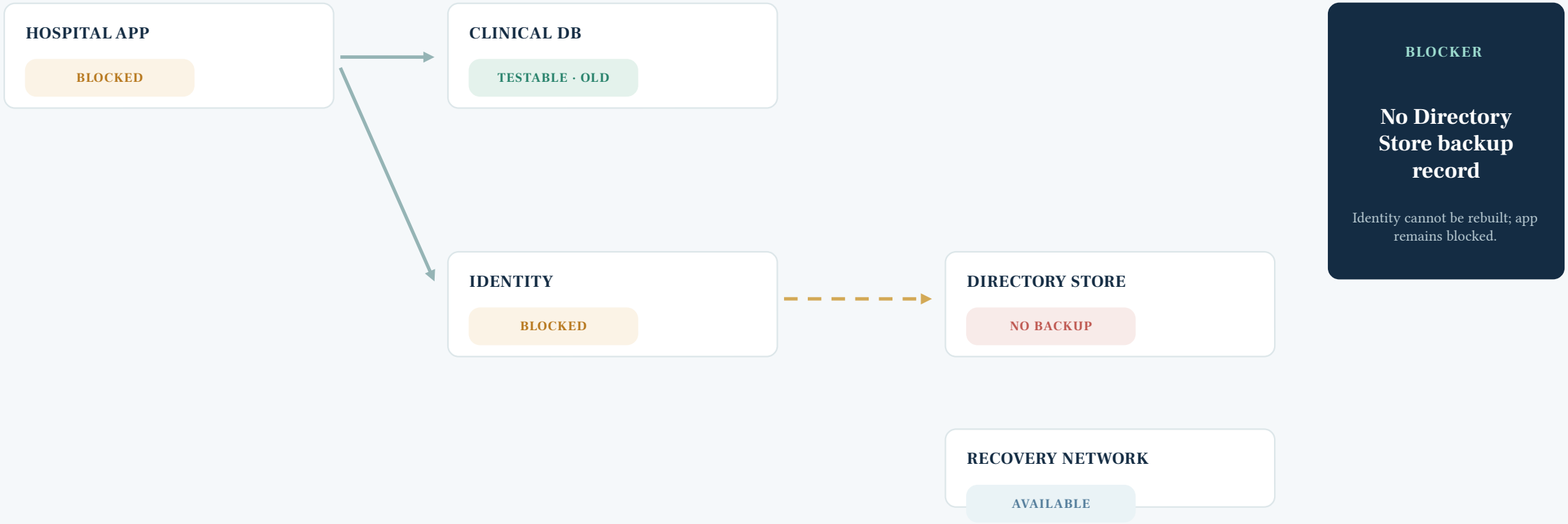
DOMAIN FILTERED

For each: Is a copy reachable? Is it within the data-loss target? What else is missing?

SERVICE DEPENDENCIES

Restore prerequisites before the hospital app

The app needs its database and identity service; identity needs directory data.



RULE-BASED RECOVERY ORDER

Example at 08:00: the 05:00 database file is one hour beyond its two-hour data-loss target.

AI explains evidence; a person controls action

Fixed rules decide what may be tested. The optional model cannot approve or run a restore.

OPTIONAL LOCAL MODEL

Evidence in

Scenario findings
permitted evidence IDs
rule-only baseline plan

Ollama 0.40.11 · Qwen3 8B

Local and optional · Apache-2.0 model

01 **Use only supplied facts**

Do not invent recovery details

02 **Cite evidence references**

Reject unknown references

03 **Say what is unknown**

A matching fingerprint is not a malware scan

04 **Wait for a person**

The person approves or rejects a test copy

DETERMINISTIC FALLBACK WHEN MODEL IS OFFLINE

No model execution occurred in the measured evaluation.

ISOLATED TEST

A real sample file copy is measured and checked

A person approved the copy; the source stayed unchanged; the test folder is separate.

RESTORE OBSERVATION

0.581 ms

mean over 3 actual local copy runs

Scope

file copy ✦ fingerprint ✦ sample data structure check

Not service recovery time

Source file	clinical database sample · West location	VERIFIED
Approval	person approved the pending test	RECORDED
Copy target	unique isolated test folder	SEPARATE VOLUME
File check	source and copy fingerprints match	MATCHED
Data check	expected JSON fields and record count	PASSED
Malware status	no scanner executed	UNKNOWN

EVALUATION

Measured against known synthetic truth

Automated tests check the explicit rules; the experiment is intentionally small and reproducible.

PYTEST SUITE

10/10

0 failures

FILE FINGERPRINT

5/5 · 100%

expected match / mismatch

BACKUP AGE

5/5 · 100%

age vs data-loss target

SERVICE BLOCKERS

12/12 · 100%

4 services × 3 scenarios

APPROVED TEST COPY

3/3 · 100%

copy + file/data checks

UNSAFE COPY DENIED

2/2 · 100%

pending or untrusted

AI BASELINE Fallback actions match rule-only baseline · evidence/action IDs constrained · local model accuracy not run

npm audit: 0 findings after image-size override · pip-audit: 0 findings · synthetic only

COMMERCIAL MODEL

Services-led pilots can fund a recurring assurance layer

Illustrative first-year planning model · all figures are assumptions, not market pricing.

CUSTOMERS	REVENUE	DIRECT COST	GROSS PROFIT*	NET RESULT*
1	$78k	$18k	$60k	**-$200k**
5	$390k	$90k	$300k	**-$60k**
10	$780k	$180k	$600k	**$80k**
25	$1.95m	$450k	$1.50m	**$750k**

Per customer first year: $18k implementation ✦ $48k license ✦ $12k support = $78k revenue.

*Gross profit = revenue minus direct delivery costs. Net result also subtracts assumed fixed business overhead. Pre-tax; excludes tax, financing, and customer acquisition/churn costs.

Measure customer ROI with validated outage costs, recovery labor and evidence-preparation hours; do not use generic loss claims.

WHAT WE NEED NEXT

Pilot the evidence layer with a critical-service team.

01	**One service catalog**	3–5 services, owners, data-loss/time-to-recover targets and dependencies
02	**One joint exercise**	Physical loss + ransomware + a compound disruption scenario
03	**One measured result**	Known gaps, approved isolated test, observed duration and report